Cement Industry: The Case of Akçansa Cement Plant

GHG Emissions and Climate Change Risk Assessment Report

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1. Introduction

1.1 Background and Rationale

The cement industry is the very important component of global infrastructure and economic development. Cement is the primary binding material for concrete, the world's most widely used construction substance. Cement plants are typically large, capital-intensive facilities, and Calcium carbonate is the primary raw material of cement. Therefore, cement plants are located near limestone quarries.

The process of producing cement involves heating limestone and clay in rotary kilns at very high temperatures to form clinker, which is then ground with gypsum to make cement. Cement production is one of the world's most carbon intensive industrial processes. It generates approximately 2.1-2.5 Gt of CO_2 emissions annually worldwide, 5% to 8% of global GHG emissions. Nearly 60% of these emissions come from the calcination of limestone ($CaCO_3 \rightarrow CaO + CO_2$), the primary chemical reaction in clinker production, while the remaining 40% comes from the combustion of fossil fuels used to reach kiln temperatures of 1400-1500°C.

Despite significant improvements in energy efficiency, such as switching from wet kilns to preheated dry kilns, GHG emissions have not decreased proportionally relative to cement production. Global cement demand continues to grow at a faster rate than the population and GDP, driven by urbanization and infrastructure development.

Cement and concrete are affordable and durable materials; however, their extensive use comes at an important climate cost. The IPCC classifies cement as a "hard-to-abate" sector, where mitigation pathways are limited but crucial for global decarbonisation. The world produced over 4.1 billion tonnes of cement in 2023, releasing nearly 0.9 tonnes of CO_2 per tonne of cement produced. As mentioned earlier the sector's total contribution to global CO_2 emissions is 5% to 8%, making it a key focus of industrial climate risk assessments. Turkey is the world's fifth largest cement producer, according to the USGS (United States Geological Survey) Mineral Commodity Summary 2023 report, producing over 70 million tonnes of cement each year and also exporting around 20 million tonnes.

The cement industry is responsible for approximately 5% of Turkey's industrial CO_2 emissions. After signing the Paris Agreement in 2016, Turkey set a Net Zero target for 2053 and introduced MRV regulations in 2017 for high emission industries such as cement. Additionally, Turkish cement companies face transition risks from the EU ETS (Emissions Trading System) and the CBAM (Carbon Border Adjustment Mechanism) as well as physical climate risks such as water shortages and heat stress, which can interrupt some operations. Due to these pressures, decarbonization strategies in Turkey's cement sector are increasingly aligned with EU sustainability reporting standards such as the TCFD (Task Force on Climate-related Financial Disclosures) and ESG (Environmental, Social, Governance).

1.2 IPCC Methodology

According to the IPCC Good Practice Guidance and Guidelines for National GHG Inventories (1996, 2006), CO_2 emissions from cement production are best estimated using clinker data rather than cement data, since the process emissions occur during clinker formation. The standard calculation is: ECO2=Clinker Production (t)×0.507×(1+CKD%)

Where 0.507 t CO_2/t clinker corresponds to the default emission factor derived from 64.6% lime content in clinker, and CKD (Cement Kiln Dust) losses contribute an additional 2–6%. This methodological framework ensures comparability of national inventories and supports plant level Monitoring, Reporting, and Verification (MRV) systems required under the UNFCCC. It forms the technical foundation for the GHG assessment in this project.

1.3 Case Study : Akçansa Çimento Sanayi A.Ş.

To bring the analysis to real world data, this project includes a facility level case study of Akçansa Çimento Sanayi A.Ş. , a leading Turkish producer jointly owned by Heidelberg Materials and Sabancı Holding. Akçansa publishes detailed sustainability reports that include Scope 1 and Scope 2 emissions, energy consumption, and alternative fuel ratios. In 2023, Akçansa's carbon intensity averaged 620 kg CO_2 per tonne of cement, below the national sector mean (670 kg CO_2/t), largely due to the use of 18% alternative fuels and waste heat recovery systems. According to Akçansa Sustainability Report 2023, the company targets continued reductions through low clinker cement formulations, renewable electricity procurement, and participation in carbon market readiness programs.

This GHG and Climate Change Risk Assessment Report aims to measure Akçansa's greenhouse gas emissions using IPCC compliant methodologies within the Scopes 1, 2, and 3; identifying the key physical and transition risks that climate change effects to cement production; evaluate mitigation strategies such as improving energy efficiency, reducing clinker content, assessing carbon capture and storage (CCS) feasibility and improving water and energy resilience through adaptation standards. Finally, to develop a replicable climate risk assessment framework for Turkey's cement industry that aligns with leading international standards and guidance, including the IPCC, IEA, and TCFD guidance.

1.4 Expected Outcomes

The study will deliver a plant specific GHG inventory and some strategic recommendations aligned with Turkey's 2053 Net Zero target and the EU Green Deal. By combining IPCC methodologies with real industrial data, the project demonstrates how quantitative GHG accounting can guide decarbonization, risk management, and policy design in the cement sector.

2. GHG Assesment

This section presents a detailed assessment of greenhouse gas (GHG) emissions for Akçansa Çimento Sanayi A.Ş. by analysing multi-year emissions data reported since 2019. The assessment follows the GHG Protocol framework and is consistent with IPCC-aligned Monitoring, Reporting, and Verification (MRV) principles. Emissions are evaluated under Scope 1, Scope 2, and Scope 3, with particular emphasis on trends observed between 2019 and 2024.

2.1 Current Situation and Average GHG Emissions (Scope 1, Scope 2, Scope 3)

Akçansa's emissions profile is dominated by direct emissions (Scope 1), which is characteristic of the cement industry due to the intrinsic carbon intensity of clinker production. Between 2019 and 2021, Scope 1 emissions remained relatively stable, fluctuating between approximately 5.68 and 5.76 million tonnes of CO_2e. This stability indicates that, despite annual variations in production volume and fuel mix, the fundamental structure of emissions did not change significantly during this period.

From 2022 onwards, a gradual decline in Scope 1 emissions is observed, reaching a minimum of approximately 5.31 million tonnes CO_2e in 2023. This reduction can be associated with operational efficiency improvements, increased use of alternative fuels, and clinker substitution strategies reported by the company. However, in 2024, Scope 1 emissions increased again to approximately 5.48 million tonnes CO_2e, suggesting a rebound effect likely linked to higher production levels and demand recovery. This pattern illustrates the strong coupling between cement output and direct process emissions.

Scope 2 emissions, which represent indirect emissions from purchased electricity, have been consistently lower in magnitude compared to Scope 1. For the years where full data is available (2022–2024), Scope 2 emissions show a modest but clear downward trend, decreasing from approximately 302 thousand tonnes CO_2e in 2022 to 293 thousand tonnes CO_2e in 2024. This decline reflects improvements in electricity efficiency, as well as partial integration of lower-carbon electricity sources. Although Scope 2 emissions account for a relatively small share of total emissions, they remain important from a transition risk perspective due to potential exposure to rising grid emission factors and carbon pricing mechanisms.

Scope 3 emissions represent a significant and growing component of Akçansa's overall carbon footprint. In 2022, Scope 3 emissions were reported at approximately 1.46 million tonnes CO_2e. This value increased substantially in 2023 and 2024, reaching approximately 1.72 and 1.76 million tonnes CO_2e, respectively. The upward trend in Scope 3 emissions highlights the increasing relevance of value chain impacts, including upstream raw material supply, fuel procurement, and downstream transportation of cement products. Unlike Scope 1 and Scope 2 emissions, Scope 3 emissions are less directly controllable and require supplier engagement and logistics optimization strategies.

2.2 Operational Phase GHG Emissions (Scope 1, Scope 2, Scope 3)

During the operational phase, GHG emissions are generated continuously as part of clinker and cement production activities. Scope 1 emissions remain the dominant source throughout all years analysed, consistently accounting for the majority of total emissions. This confirms that process-related CO_2 from limestone calcination remains the principal decarbonisation challenge for Akçansa, as is the case across the global cement sector.

Operational Scope 2 emissions demonstrate a relatively stable but gradually improving performance. The declining trend observed after 2022 indicates that efficiency measures in grinding and auxiliary processes have been effective in limiting electricity related emissions, despite variations in production intensity. These reductions contribute positively to overall emissions management but do not fundamentally alter the plant's carbon profile.

Operational Scope 3 emissions have increased in recent years, suggesting that indirect emissions outside the plant boundary are becoming more prominent as direct emissions are gradually optimised. This shift implies that future emission reduction strategies will increasingly need to extend beyond plant level technologies and address supply chain and distribution systems. From an operational risk perspective, rising Scope 3 emissions also increase exposure to future regulatory requirements related to value chain reporting and carbon border mechanisms.

2.3 Total GHG Emissions

Total GHG emissions, calculated as the sum of Scope 1, Scope 2, and Scope 3 emissions, provide a comprehensive view of Akçansa's climate impact. For the years with complete datasets (2022–2024), total emissions decreased from approximately 7.13 million tonnes CO_2e in 2022 to

7.03 million tonnes CO_2e in 2023, followed by an increase to approximately 7.24 million tonnes CO_2e in 2024. This pattern reflects the combined effect of operational efficiency gains and fluctuations in production volume.

The temporary reduction observed in 2023 suggests that emission mitigation measures can deliver measurable improvements in total emissions. However, the subsequent increase in 2024 demonstrates that without structural changes such as deep clinker substitution or carbon capture technologies absolute emission reductions remain vulnerable to production growth. Overall, the emissions trend confirms that while Akçansa performs relatively well compared to sector averages, its total GHG emissions remain strongly influenced by the inherent carbon intensity of cement manufacturing.

Year	Scope 1 (tCO_2e)	Scope 2 (tCO_2e)	Scope 3 (tCO_2e)	Total GHGs (tCO_2e)
2019	5,688,037	—	—	—
2020	5,680,999	—	—	—
2021	5,764,763	—	—	—
2022	5,533,084	301,672	1,457,047	7,128,201
2023	5,309,000	300,057	1,718,887	7,027,887
2024	5,484,834	292,562	1,758,743	7,243,577

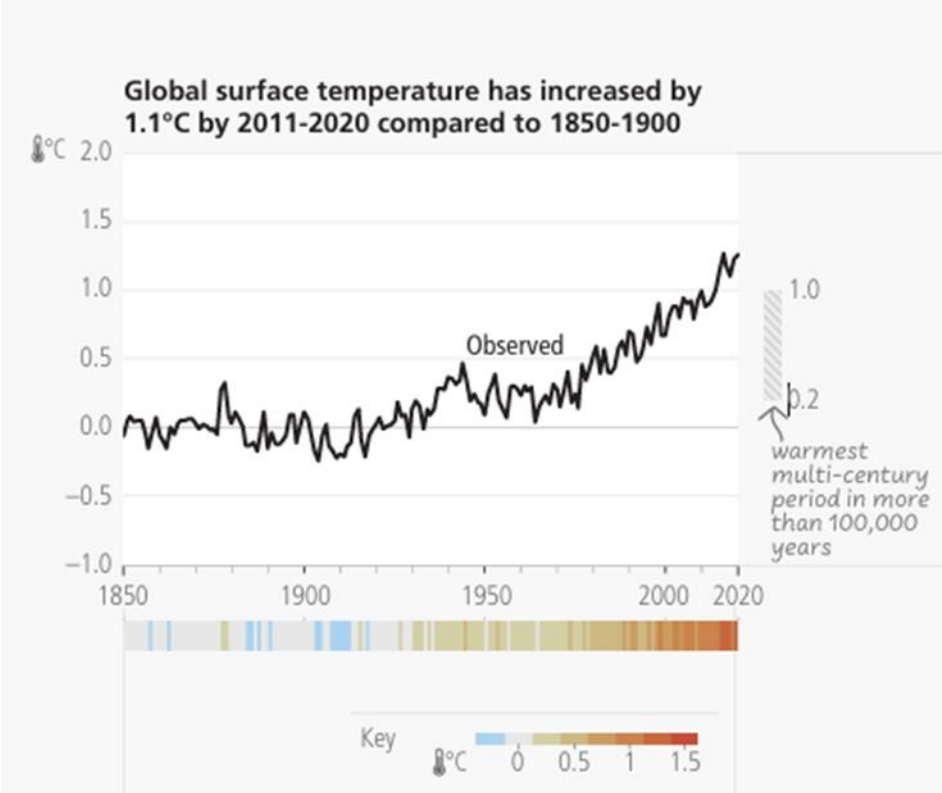
Impacts of the 6 February 2023 Earthquakes on the Cement Sector

The earthquakes of 6 February 2023 had notable impacts on Turkey's cement sector by causing temporary production disruptions and logistical constraints in the affected regions. Damage to infrastructure and energy supply led to short-term operational interruptions at several cement plants, highlighting the sector's exposure to acute physical risks. In the subsequent reconstruction phase, cement demand increased significantly, resulting in higher production levels and associated GHG emissions across the sector. This dual effect demonstrates how extreme events can both disrupt industrial operations in the short term and intensify emission pressures in the long term through reconstruction driven demand.

3. Climate Change Risk Assesment

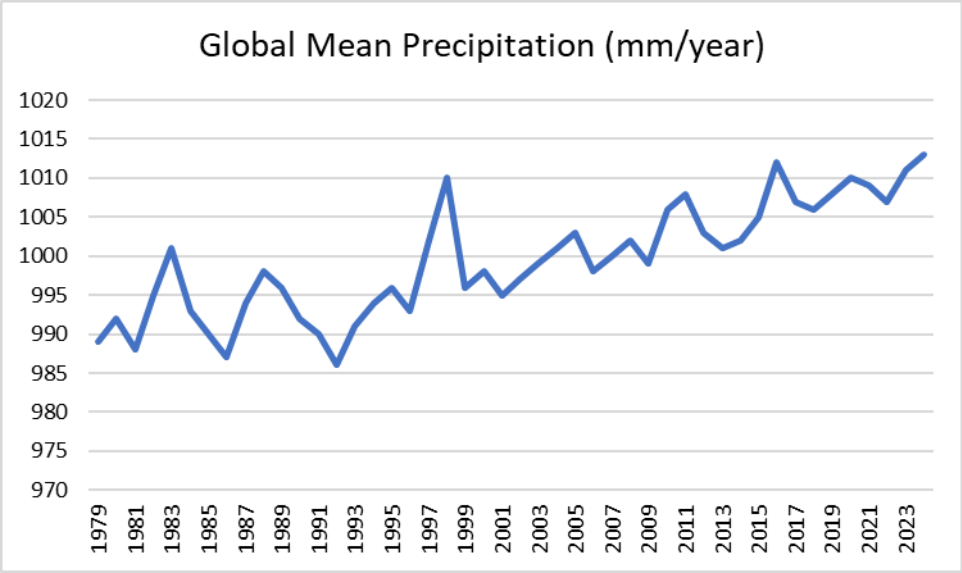
3.1 Observation

3.1.1 Temperature



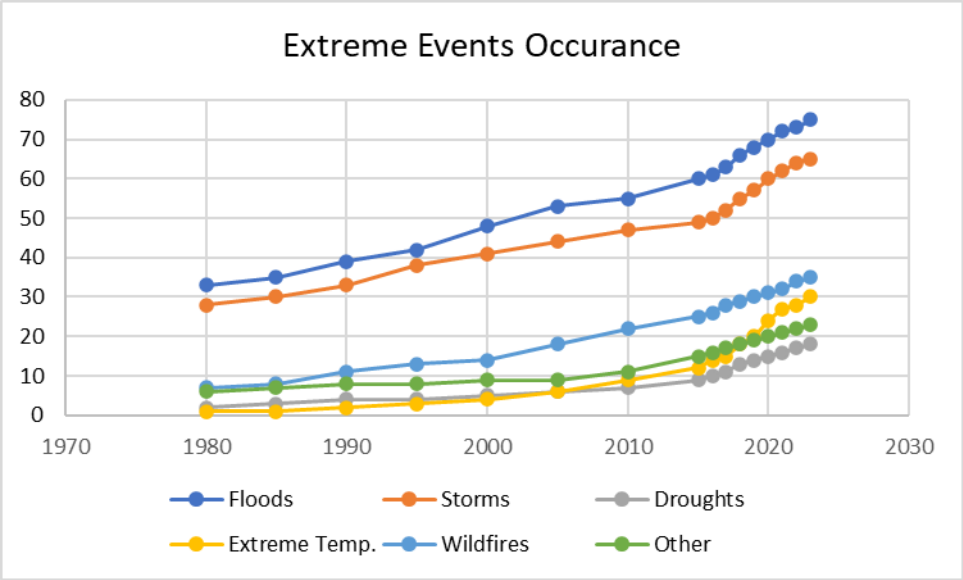
The temperature graph shows a clear and sustained increase in global mean surface temperature since the late 19th century. Relative to the 1850–1900 pre-industrial baseline, global surface temperature increased by approximately 1.1°C during the 2011–2020 period, as reported by the IPCC AR6 Working Group I. The curve indicates relatively small interannual variability until the mid-20th century, followed by a pronounced acceleration after 1950, and an even steeper rise after 1980. By around 2020, annual temperature anomalies exceed +1.0°C, placing the current climate close to the +1.5°C global warming level discussed extensively in IPCC assessments. According to IPCC AR6, the rate of warming observed since 1970 is higher than in any comparable period over at least the last 2,000 years, confirming that the recent warming trend is unprecedented in the context of human civilization.

3.1.2 Global Mean Precipitation



The precipitation graph presents global mean annual precipitation values from 1979 to 2023, ranging approximately between 990 mm/year and 1,015 mm/year. Over this period, a modest upward trend is observable, with global mean precipitation increasing by roughly 15–20 mm/year between the early 1980s and the early 2020s, corresponding to an increase of about 1.5–2%. However, the graph also shows substantial interannual variability, with pronounced peaks and troughs rather than a smooth trend. This behavior is consistent with IPCC AR6 conclusions, which state that global warming leads to only small increases in total global precipitation (about 1–3% per °C of warming), while much larger changes occur in precipitation intensity and temporal distribution. Therefore, the relatively small numerical increase in global mean precipitation masks significant regional and event-scale hydrological changes.

3.1.3 Extreme Climate Events



The extreme events graph illustrates a strong upward trend in the number of climate-related disasters reported globally since 1980, including floods, storms, droughts, extreme temperature events, and wildfires. Flood events increase from approximately 30–35 events per year in the early 1980s to over 70 events per year by the early 2020s, making floods the most frequently recorded climate-related hazard. Storm events show a similar pattern, rising from around 25–30 events per year to more than 60 events per year. Extreme temperature events exhibit the fastest relative growth, increasing from fewer than 5 events per year in the 1980s to approximately 30 events per year by 2023. According to IPCC AR6, the increase in hot extremes and heatwaves is assessed with high to very high confidence and is directly attributable to anthropogenic climate change.

3.1.4 Drought and Long-Term Water Stress

Although drought events appear less frequent than floods or storms in the disaster statistics rising from roughly 2–3 events per year in the 1980s to about 15–20 events per year by the 2020s this does not indicate a lower level of risk. IPCC AR6 emphasizes that drought is a slow onset hazard, which is often underrepresented in disaster databases. Importantly, rising temperatures increase potential evapotranspiration, meaning that soils and vegetation lose moisture more rapidly even when total annual precipitation does not decline significantly. As a result, regions may experience simultaneous increases in extreme rainfall events and longer dry periods, a phenomenon known as hydro climatic intensification. The combination of higher temperatures, modest precipitation increases, and greater rainfall variability therefore leads to an overall

increase in agricultural, ecological, and hydrological drought risk, despite the relatively smaller number of reported drought disasters.

3.1.5 Conclusion

Taken together, the temperature, precipitation, and extreme-event graphs provide a consistent and physically coherent picture of ongoing climate change. Global mean temperature has already increased by over 1.1°C, driving modest increases in global mean precipitation, but much stronger increases in climate extremes, particularly floods, storms, heatwaves, and wildfires. The numerical trends observed in the graphs align closely with IPCC AR6 conclusions that climate change primarily manifests through the intensification of extremes rather than large changes in annual means. These observed changes form a robust empirical basis for assessing future physical climate risks, particularly for water-intensive and energy-intensive industrial sectors.

3.2 Projection

3.2.1 Projected global temperature change

IPCC AR6 projects that global surface temperature (GSAT) continues to increase across all scenarios in the near term, with widening divergence after mid century driven by cumulative emissions.

Scenario	2021–2040	2041–2060	2081–2100
SSP1-1.9	1.4 (1.2–1.7)	1.6 (1.3–1.9)	1.4 (1.0–1.8)
SSP1-2.6	1.4 (1.2–1.7)	1.8 (1.5–2.1)	2.0 (1.3–2.7)
SSP2-4.5	1.5 (1.3–1.8)	2.1 (1.7–2.5)	2.7 (2.1–3.5)
SSP3-7.0	1.6 (1.3–1.9)	2.4 (2.0–2.8)	3.6 (2.8–4.6)
SSP5-8.5	1.6 (1.3–1.9)	2.4 (2.0–2.8)	4.4 (3.3–5.7)

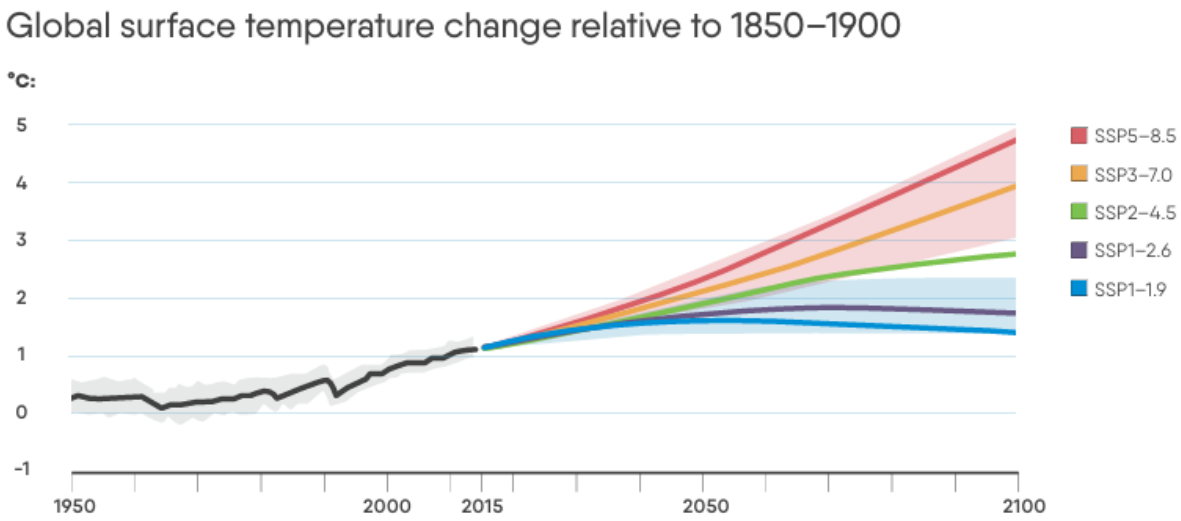


Figure 3: IPCC AR6 WGI Figure SPM.8a

This table shows IPCC AR6 projections of global mean surface temperature change relative to the 1850–1900 baseline under different Shared Socioeconomic Pathways (SSPs). In the near term (2021–2040), warming levels are similar across all scenarios, reaching approximately 1.4–1.6°C, reflecting committed warming from past emissions. After mid-century, scenarios diverge significantly. Under strong mitigation (SSP1-1.9), global warming stabilizes around 1.4–1.6°C by 2100, consistent with the 1.5°C target. In contrast, intermediate pathways such as SSP2-4.5 lead to about 2.7°C warming by late century, while high-emission scenarios (SSP5-8.5) result in severe warming of approximately 4.4°C by 2081–2100. These differences imply substantially higher risks of heat extremes, hydrological stress, and compound climate hazards under higher-emission pathways.

3.2.2 Projected changes in precipitation

At the global scale, warming intensifies the hydrological cycle: wet regions and wet seasons tend to get wetter, while many subtropical land regions face drying tendencies, and extreme rainfall intensifies strongly. For risk assessment, extremes are often more decision relevant than mean rainfall because they drive flood, drainage, and infrastructure thresholds.

Global warming level	Frequency (times per decade)	Intensity change
Present (~1°C)	1.3 (1.2–1.4)	+6.7%

1.5°C	1.5 (1.4–1.7)	+10.5%
2°C	1.7 (1.6–2.0)	+14.0%
4°C	2.7 (2.3–3.6)	+30.2%

This table shows extreme precipitation (global land, 1-day “10-year event”) rises in both frequency and intensity with warming.

3.2.3 Projected Extreme Event Changes

Extreme Event Type	Projected Change (2020s–2100)	Confidence Level (IPCC AR6)
Floods	Strong increase in frequency and intensity	High
Storms	Increase in intensity and rainfall	Medium–High
Droughts	Increased frequency and severity	Medium
Extreme Temperature	Very strong increase in frequency and intensity	Very High
Wildfires	Increased fire weather and burned area	Medium
Other (Compound events)	Increased occurrence and impact	Medium–High

IPCC AR6 projections indicate that climate-related extreme events intensify substantially between the 2020s and 2100. Floods and extreme precipitation events increase in both frequency and intensity, storms become more damaging due to stronger rainfall and winds, droughts intensify due to enhanced evapotranspiration, and extreme heat events show the strongest increase, becoming near-annual occurrences under high warming scenarios. Wildfire risk and compound extreme events further amplify climate related impacts, highlighting the growing importance of multi-hazard risk assessment.

3.3 Risks

Climate Risk Matrix with Impact, Vulnerability and Measures

Hazard /	Impact	Vulnerability	Measures (Risk	~1.5°C	~2.0°C	~3.0°C	~4.0°C

Impact Area	(What happens?)	(Who/what is exposed?)	reduction options)				
Extreme Heat	Increased frequency and intensity of heatwaves; heat stress	Human health, workers, energy systems, ecosystems	Heat action plans, cooling systems, operational scheduling	High	Very High	Very High	Very High
Floods & Extreme Precipitation	More intense rainfall, river and flash floods	Urban areas, infrastructure, industrial facilities	Drainage upgrades, flood defenses, site protection	Medium	High	Very High	Very High
Storms	Stronger storms, higher rainfall and wind damage	Coastal regions, transport, buildings	Structural reinforcement, early warning systems	Medium	High	High	Very High
Drought & Water Scarcity	Reduced soil moisture, water shortages	Agriculture, industry, ecosystems, water supply	Water efficiency, alternative water sources	Medium	High	Very High	Very High
Wildfires	Longer fire seasons, higher fire intensity	Forests, rural settlements, air quality	Fuel management, firebreaks, emergency planning	Medium	High	Very High	Very High
Food Production	Declining crop yields and fisheries productivity	Food systems, rural livelihoods	Climate-resilient crops, diversification	Medium	High	Very High	Very High
Biodiversity Loss	Species loss, ecosystem degradation	Terrestrial and marine ecosystems	Conservation, habitat restoration	High	Very High	Very High	Very High
Human Health (Heat–Humidity)	Increased mortality and morbidity	Elderly, outdoor workers, vulnerable populations	Health system adaptation, heat-health warning systems	Medium	High	Very High	Very High
Coastal Systems & Sea Level Rise	Coastal flooding, erosion, salinization	Coastal cities, deltas, infrastructure	Coastal defenses, managed retreat	Medium	High	Very High	Very High
Compound Events	Combined hazards (e.g. heat–drought–wildfire)	All socio-economic systems	Integrated risk management, cross-sector planning	Medium	High	Very High	Very High

The risk matrix integrates projected climate hazards with their associated impacts,

vulnerabilities, and potential risk reduction measures. It demonstrates that climate-related risks increase rapidly with rising global warming levels. At approximately +1.5°C, several sectors already face high risks, while beyond +2°C most hazards transition from medium to high risk. At warming levels of +3°C and above, risks become very high across nearly all systems, reflecting widespread, severe, and potentially irreversible impacts, consistent with IPCC AR6 assessments.

4. Conclusion

This report demonstrates that current global greenhouse gas (GHG) emission trajectories are not aligned with pathways required to limit global warming to 1.5°C or 2°C. As illustrated by IPCC scenarios, implemented policies and existing national commitments (NDCs) lead to continued high emissions, resulting in warming levels of 3°C or higher by 2100. In contrast, limiting warming requires rapid and sustained emission reductions and the achievement of net zero emissions by mid century.

The analysis confirms that climate risks increase sharply with each increment of global warming. At around 1.5°C, risks related to extreme heat and biodiversity loss are already high. As warming approaches 2°C and beyond, flood, drought, food production, and human health risks escalate rapidly. Under higher warming scenarios, these risks become very high, widespread, and potentially irreversible, particularly due to the increasing frequency of extreme and compound events.

Overall, the findings highlight that early and ambitious GHG mitigation is a key risk-reduction strategy. Delayed action significantly increases the severity of climate impacts, whereas limiting emissions reduces long-term climate risks and supports a more manageable and resilient future.

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